

PROJECT REPORT
On
[AI-Based Weather Advisory System using Bayesian Inference]
Submitted
Of
“TAE-1(Project Based Learning)” Under the
subject
Artificial Intelligence
(VI Semester, B.Tech CSE)
(ACADEMIC SESSION 2025-26)

Criteria	Rubrics	Marks
AI Technique(s) (2M)	Selection of suitable technique	
Problem Representation (4M)	Representation of the method and knowledge required by the statement	
Tools explored (2M)	Use of AI tool from industry perspective	
Outcome (2M)	Analysis of project working	
Total Marks		

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1. Title: AI-Based Weather Advisory System using Bayesian Inference

AI-Based Weather Advisory System using Bayesian Inference This system focuses on providing intelligent, real-time weather predictions using probabilistic modeling instead of traditional fixed predictions. It leverages AI techniques to deliver confidence-based advisories, helping users make informed decisions.

2. Problem Statement:

Weather prediction is critical in sectors like agriculture, aviation, logistics, and disaster management. However, traditional weather forecasting models are mostly **deterministic**, meaning they provide fixed outputs (e.g., “it will rain”) without indicating uncertainty.

In reality, weather is **highly dynamic and uncertain**, influenced by multiple interacting factors such as humidity, pressure, temperature, and wind patterns.

These models often struggle with:

- Incomplete or noisy data
- Sudden environmental changes
- Lack of real-time adaptability

Because of this, incorrect predictions can lead to:

- Crop damage in agriculture
- Flight delays or safety risks in aviation
- Logistics inefficiencies
- Poor disaster preparedness

Therefore, there is a strong need for a system that can:

- Handle uncertainty effectively
- Continuously update predictions
- Provide **probabilistic outputs (confidence levels)** rather than binary decisions

3. Objective:

- To design and develop an AI-powered system that predicts weather conditions using probabilistic approaches instead of deterministic models

- To implement **Bayesian Inference** for updating predictions dynamically based on new incoming data
- To generate **probability-based weather advisories** (e.g., 70% rain probability, 40% storm risk)
- To improve decision-making in industries by providing **reliable and interpretable outputs**
- To enable real-time data integration from multiple sources such as APIs, IoT sensors, and satellite feeds
- To build a scalable system that can be extended to different geographical regions
- To minimize risks associated with incorrect weather predictions

4.Detail Document :

Industry Context:

Weather forecasting plays a crucial role in multiple domains:

- **Agriculture** – Farmers rely on weather forecasts for irrigation, sowing, and harvesting decisions
- **Aviation & Maritime** – Accurate weather prediction ensures safe navigation and scheduling
- **Disaster Management** – Early warnings help in reducing the impact of floods, cyclones, and storms
- **Smart Cities** – Weather data is used for traffic control, pollution monitoring, and infrastructure planning
- **Logistics & Ride-sharing** – Helps in route optimization and demand forecasting
- **Weather Apps** – Provide real-time updates to millions of users

Feasibility Study:

Weather systems are inherently uncertain, making probabilistic models highly suitable.

Why Bayesian Models?

- Can work with **incomplete and noisy datasets**
- Continuously update predictions using new evidence
- Provide **confidence levels** instead of fixed outputs
- Handle dependencies between multiple variables

Data Sources:

- Satellite imagery
- IoT-based environmental sensors
- Historical weather datasets
- Real-time weather APIs (e.g., OpenWeather, WeatherStack)

AI Technique – Bayesian Inference:

Bayesian inference updates the probability of an event based on new evidence:

Formula:

$$P(H | E) = \frac{P(E | H) \times P(H)}{P(E)}$$

Where:

H (Hypothesis): Event to predict (e.g., Rain)

E (Evidence): Observed data (humidity, temperature, wind speed)

Explanation:

Bayesian inference updates the probability of an event based on new evidence.

For example:

- If humidity is high and pressure is low → probability of rain increases
- As new data arrives → prediction gets updated dynamically



Applications:

- Rain prediction
- Storm probability estimation
- Temperature analysis
- Extreme weather risk assessment
- Air quality and pollution prediction

Model Representation:

- Uses **Directed Acyclic Graph (DAG)**
- Nodes represent variables:
 - Humidity
 - Temperature
 - Wind Speed
 - Atmospheric Pressure
 - Rain
- Edges represent dependencies:
 - Humidity → Rain
 - Pressure → Rain
 - Wind Speed → Storm

Characteristics:

- Captures relationships between variables
- Works with uncertain data
- Supports real-time updates
- Scalable and flexible

Tools & Technologies:

- **Programming Language:**
 - Python (due to strong AI/ML ecosystem)
- **Libraries:**
 - PyMC / pgmpy – Bayesian modeling
 - NumPy – Numerical computations
 - Pandas – Data handling
 - Scikit-learn – Preprocessing & evaluation

➤ **Data Sources:**

- Weather APIs (OpenWeatherMap, WeatherStack)
- IoT sensors (temperature, humidity sensors)

➤ **Infrastructure:**

- Cloud platforms (AWS, Azure, GCP)
- Big Data tools for large-scale processing

5. Proposed Solution :

The proposed system uses Bayesian inference to predict weather conditions by analyzing multiple environmental factors such as humidity, temperature, and wind speed.

Step-by-Step Workflow:

1) Data Collection

- a. Gather historical and real-time weather data
- b. Use APIs and sensors

2) Data Preprocessing

- a. Handle missing values
- b. Normalize and clean data
- c. Remove noise

3) Model Building

- a. Construct Bayesian Network
- b. Define relationships between variables

4) Model Training

- a. Train using historical datasets
- b. Learn probability distributions

5) Inference & Prediction

- a. Apply Bayesian formula
- b. Generate probability of events

6) Real-Time Updates

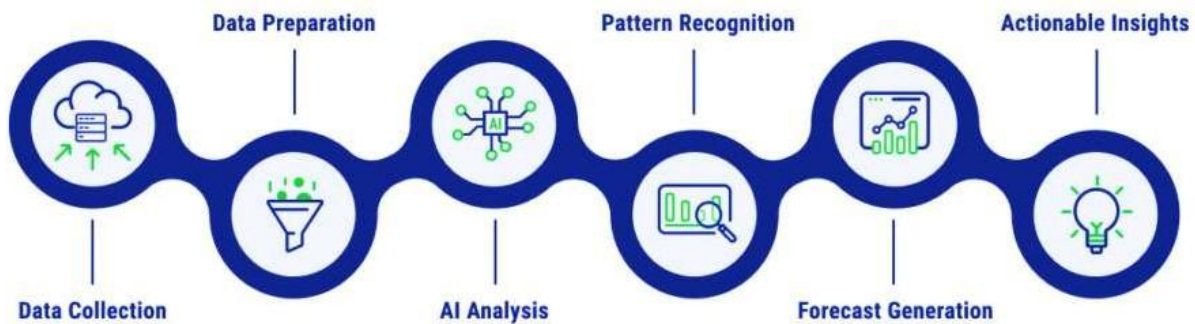
- a. Continuously update predictions as new data arrives

7) Advisory Generation

- a. **Example outputs:**
 - i. “Rain probability: 80%”
 - ii. “Storm risk: 60%”

8) User Interface / Dashboard

- a. Display results via web or mobile app
- b. Provide alerts and notifications



6.Result/Outcome :

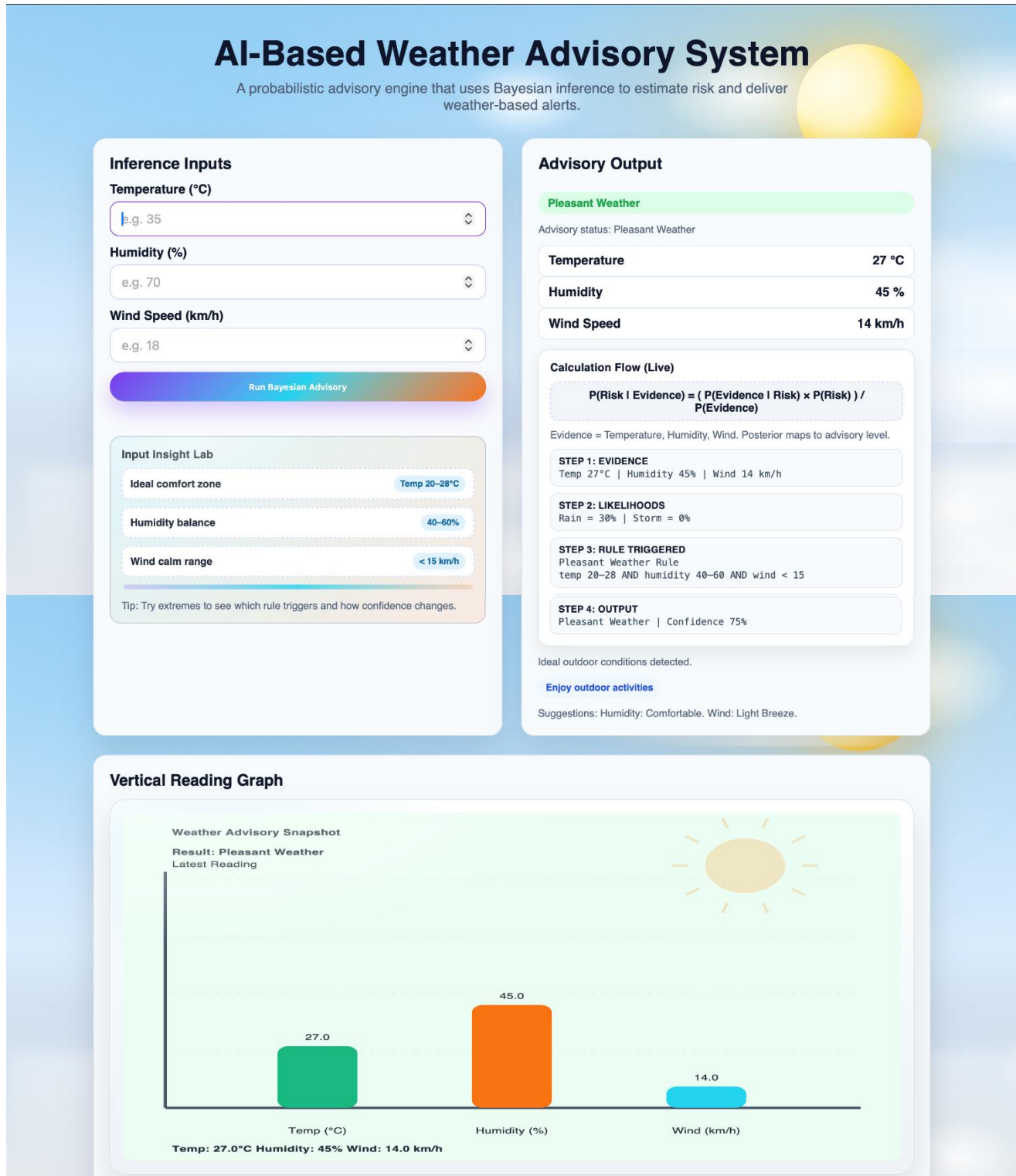
Expected Outcomes:

- More accurate and reliable weather predictions
- Real-time alerts and advisories
- Probability-based outputs instead of binary results
- Improved decision-making across industries
- Reduced operational risks and losses

Performance Metrics (Detailed)

- **Prediction Accuracy**
 - Measures correctness of predictions
- **Confidence Level**
 - Indicates certainty of predictions
- **Response Time**
 - Time taken to generate predictions
- **False Positives**
 - Predicting rain when it doesn't occur
- **False Negatives**
 - Missing actual weather events
- **Model Efficiency**
 - Computational performance and scalability

7. Screen Shots :



Note: Bar heights are scaled proportionally to each metric's value.

Storage & Graphical Summary

Total Readings

8

Latest Temperature

27.0 °C

Latest Humidity

45%

Latest Wind

14.0 km/h

Latest Result

Pleasant Weather

Save Graph Snapshot

Clear Stored History

Clear Snapshots

No saved graph snapshots yet.

Stored locally in your browser for quick analytics and offline review.

Industry Notes

Detailed theory, modeling notes, and real-world context are available on a separate page.

[View Industry Briefing](#)

AI-Based Weather Advisory System

Industry theory and modeling context for probabilistic weather advisories.

[Back to Dashboard](#)

Industry Theory

1. Industry Problem Context

In the IT industry, weather-based advisories support:

- Disaster management platforms and public alert systems
- Transportation and aviation scheduling systems
- Smart city monitoring dashboards
- Energy demand forecasting and grid operations
- Logistics and supply chain planning

The core goal is to estimate risk and recommend actions based on evolving conditions.

2. Feasibility in IT Systems

- Weather observations are modeled as structured data streams.
- Features include temperature, humidity, wind, and derived indicators.
- Data is stored in time-series databases or event pipelines.
- Real-time scoring and alerting is scalable in cloud infrastructure.

3. AI Techniques Used in Industry

- Bayesian inference for probabilistic risk estimation.
- Rule-based heuristics layered on top for safety thresholds.
- Ensembles of probabilistic models for robust decisions.

4. Representation / Modeling

- Probability model over weather events (rain, storm, heatwave).
- State = latest readings, transitions = updated sensor inputs.
- Condition rules map probabilities to advisory levels.

5. Tools & Technologies

- Python for model prototyping and data analysis.
- APIs for live weather feeds and sensor integration.
- Dashboards for visualization and alerting.

6. Industry-Level Outcome

- Real-time advisory generation with confidence scoring.
- Reduced risk via proactive warnings.
- Metrics: accuracy, false alert rate, response time.

7. Real-World Industry Use

- Public weather alert systems and emergency response.
- Airports and transport hubs for delay planning.
- Power utilities for demand and outage readiness.
- Logistics providers for route and safety decisions.

About This System

This advisory engine mirrors how IT systems use probabilistic models (like Bayesian inference) to turn live weather readings into actionable alerts. It is commonly applied in disaster response, transport scheduling, smart-city monitoring, and energy/logistics planning, where quick risk estimation improves safety and efficiency.